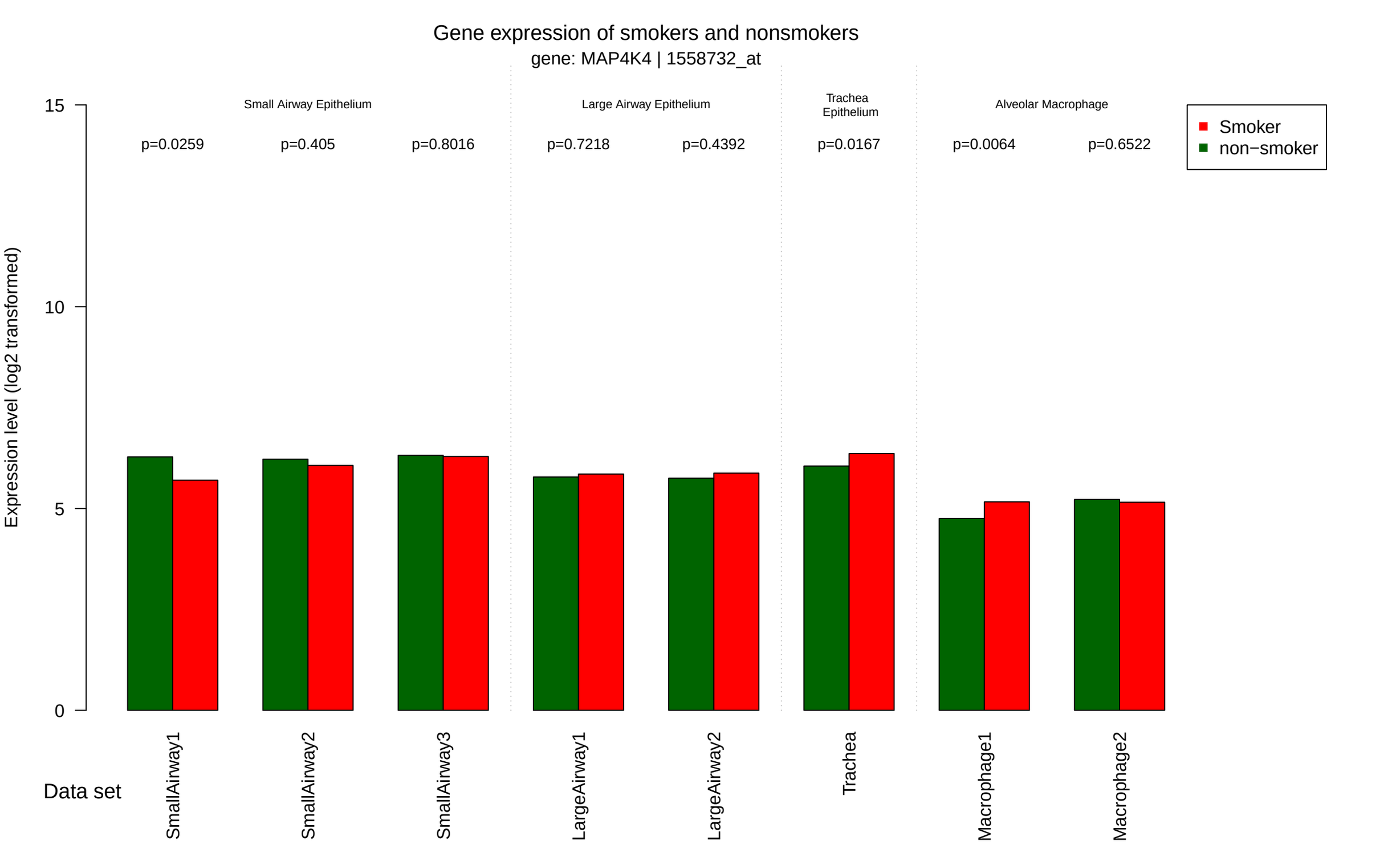
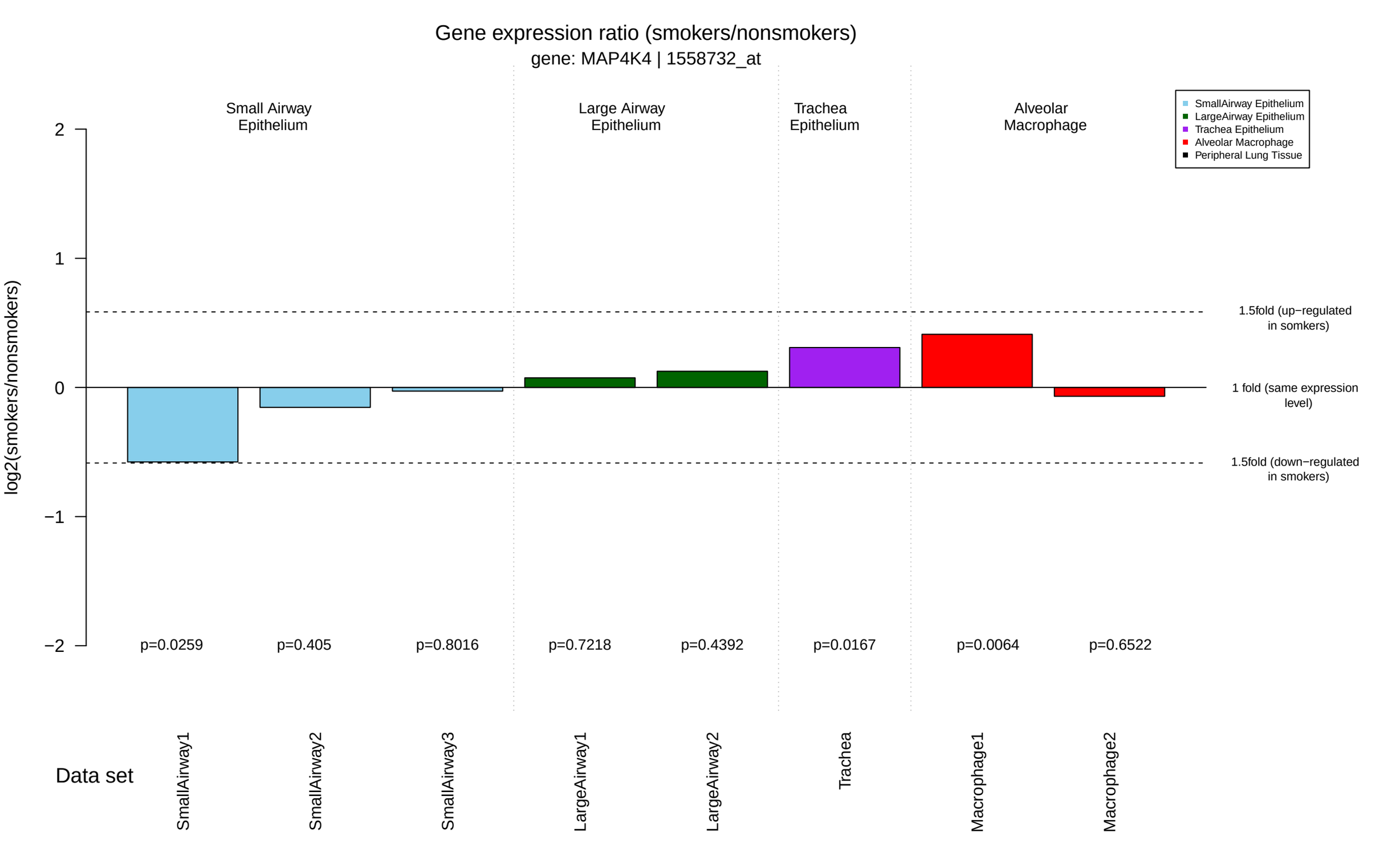
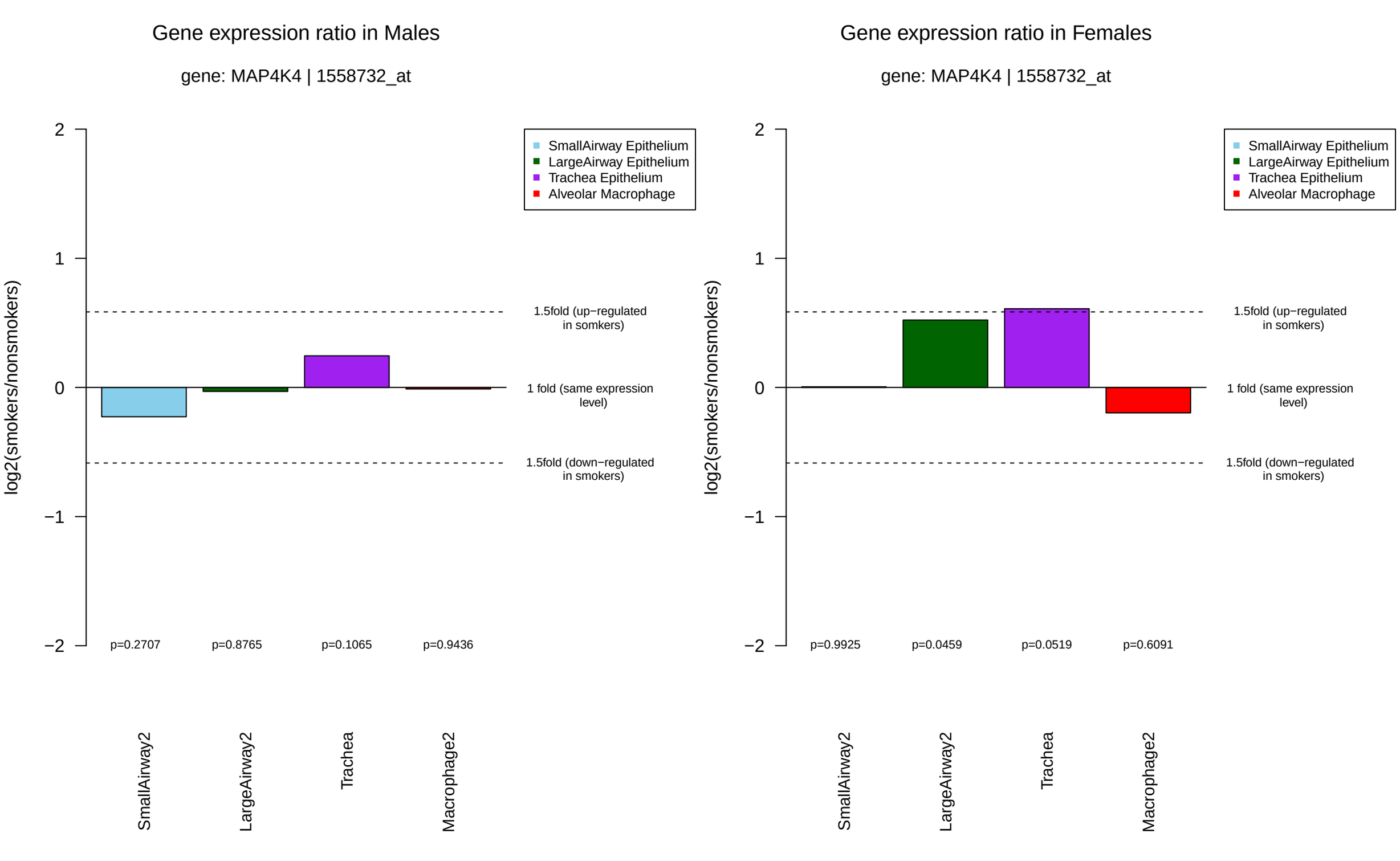
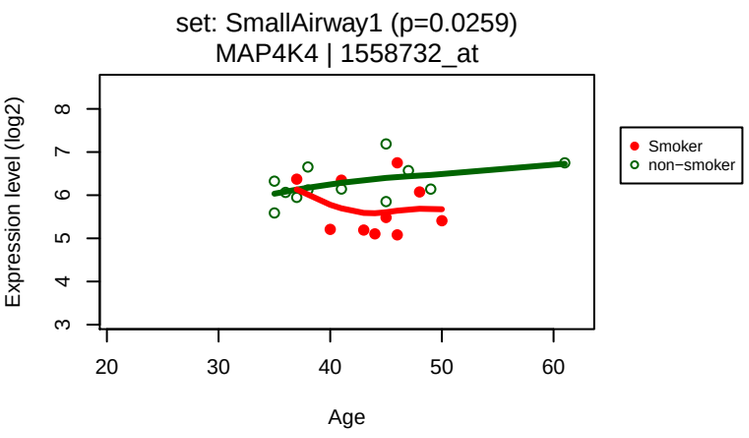
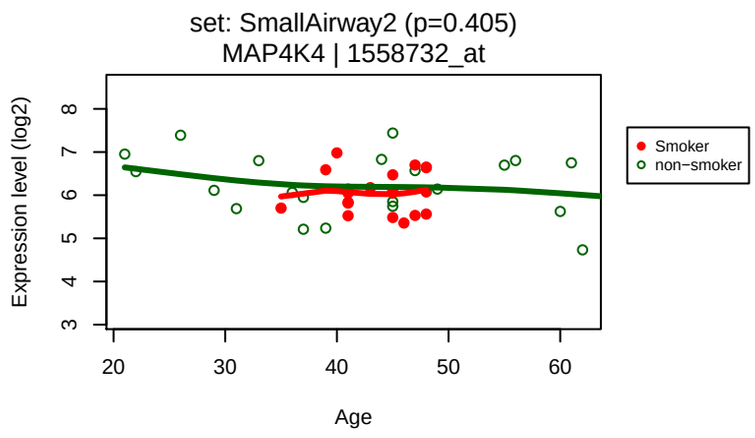
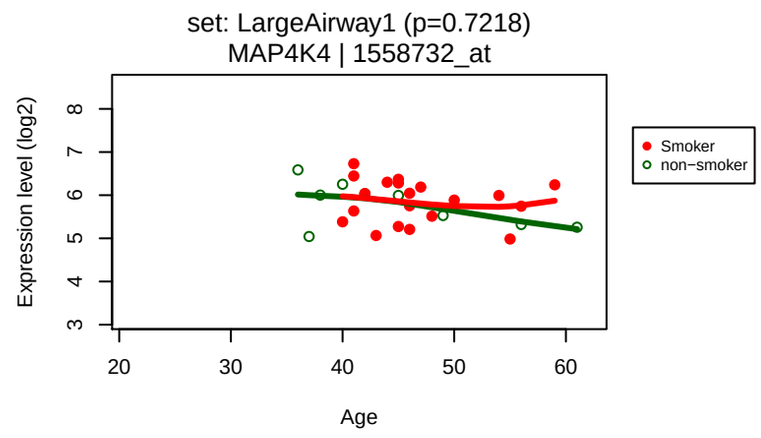

Gene expression of smokers and nonsmokers
gene: MAP4K4 | 1558732_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0259
p=0.405
p=0.8016
p=0.7218
p=0.4392
p=0.0167
p=0.0064
p=0.6522
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

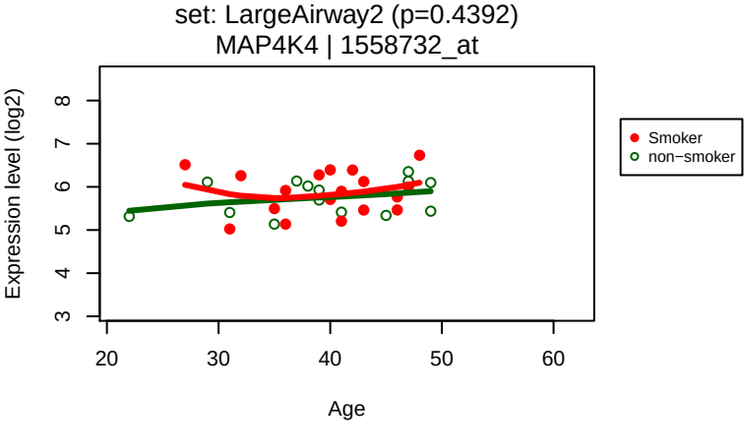

Gene expression ratio (smokers/nonsmokers)
gene: MAP4K4 | 1558732_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0259
p=0.405
p=0.8016
p=0.7218
p=0.4392
p=0.0167
p=0.0064
p=0.6522
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

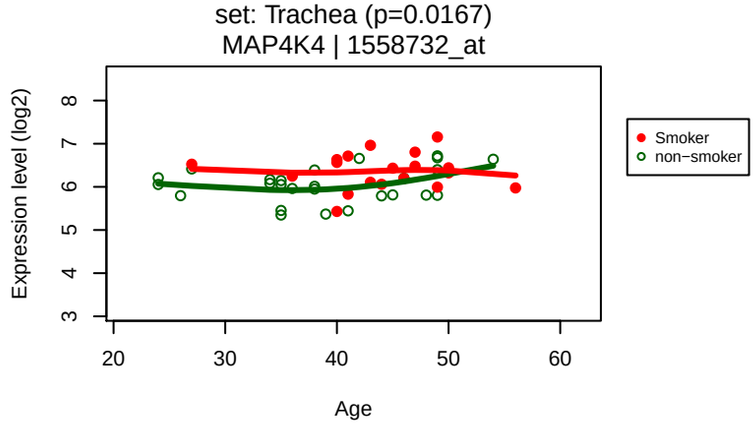

Gene expression ratio in Males
gene: MAP4K4 | 1558732_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2707
p=0.8765
p=0.1065
p=0.9436
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MAP4K4 | 1558732_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9925
p=0.0459
p=0.0519
p=0.6091
SmallAirway2
LargeAirway2
Trachea
Macrophage2

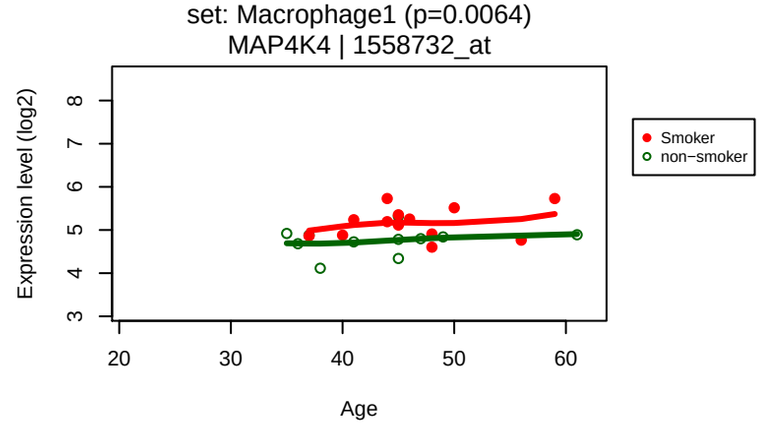

set: SmallAirway1 (p=0.0259)
MAP4K4 | 1558732_at
Expression level (log2)
Age
Smoker
non−smoker

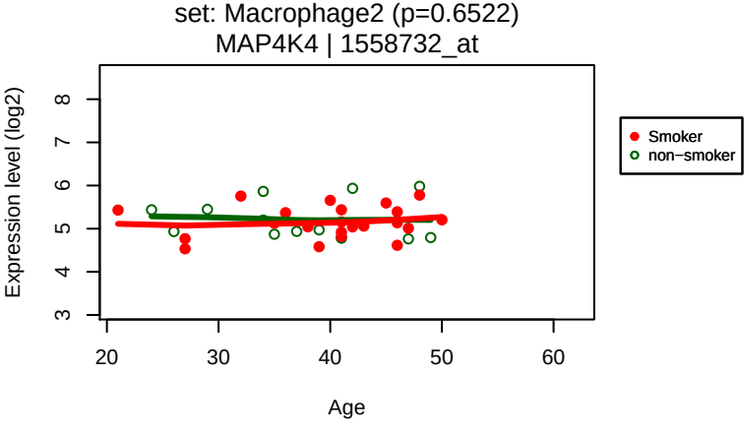

set: SmallAirway2 (p=0.405)
MAP4K4 | 1558732_at
Expression level (log2)
Age
Smoker
non−smoker

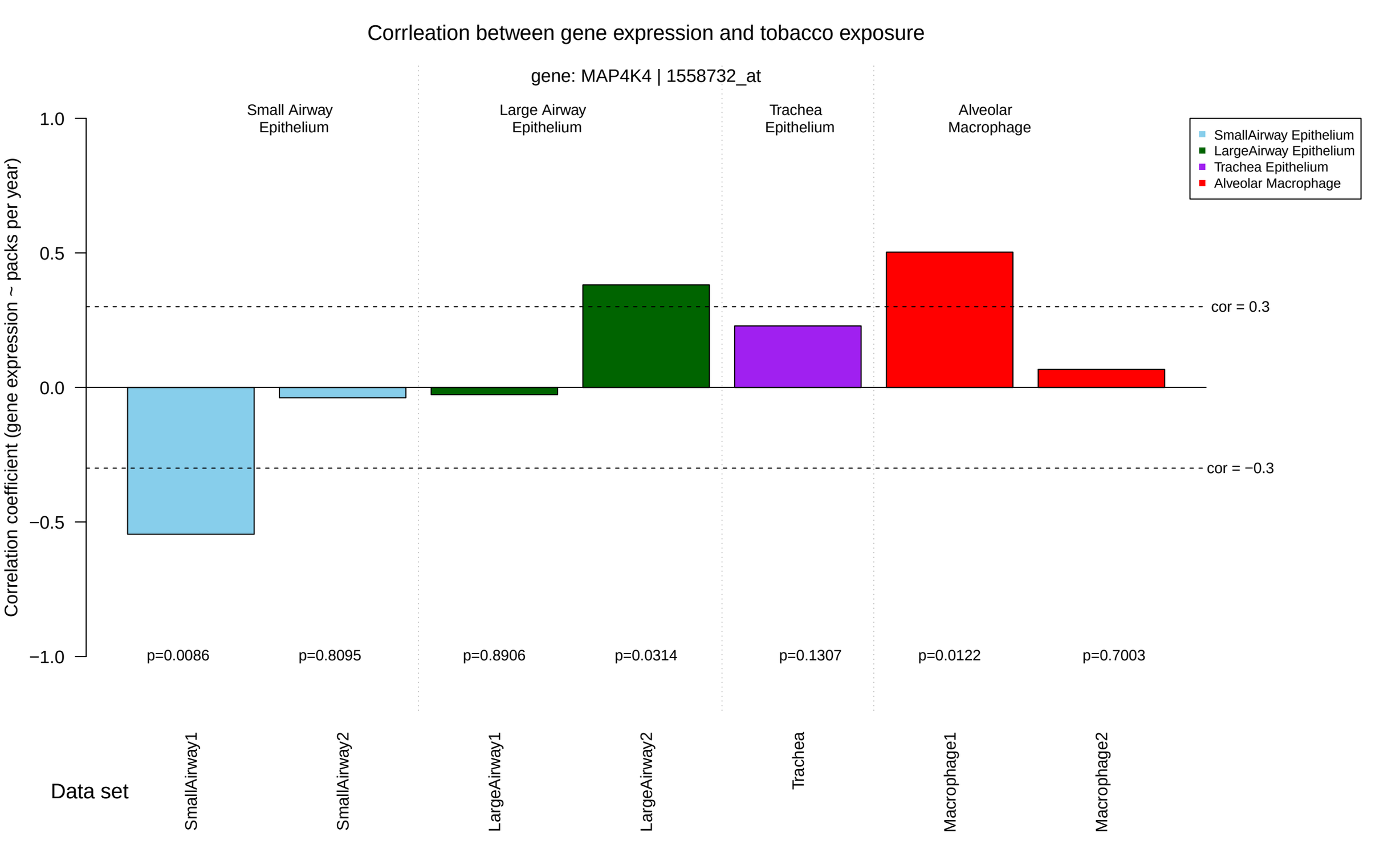

set: LargeAirway1 (p=0.7218)
MAP4K4 | 1558732_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4392)
MAP4K4 | 1558732_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0167)
MAP4K4 | 1558732_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0064)
MAP4K4 | 1558732_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6522)
MAP4K4 | 1558732_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MAP4K4 | 1558732_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0086
p=0.8095
p=0.8906
p=0.0314
p=0.1307
p=0.0122
p=0.7003
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2